

Shrey Shingala

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SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++, C, C#, Swift, Bash

Technologies: Node.js, React, FastAPI, Django, Next.js, Docker, PostgreSQL, Pandas, AWS

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Software Engineering (GPA: 4.0/4.0)

EXPERIENCE

Software Engineer Intern | Python, Django, Celery, TypeScript, React

May 2026 – Aug. 2026

Leap Tools

Toronto, ON

- Architected a marketing automation engine (Celery cronjob) compiling trigger rules into **Django Q** objects and integrating data across **4 backend services**, delivering **1,000+ emails/day** with zero duplicates
- Migrated notification handling for **10,000+** live loan records to a new automation service behind a **feature flag**, with an **idempotent cross-service backfill** keeping systems coexisting with **zero duplicate/missed sends**
- Reduced SQL query counts **80%+** and p95 latency **30%+** across **10+ endpoints** by eliminating cache-bypassing calls, profiled end-to-end with **Django Silk** and **Sentry**
- Owned production observability by building the team's **Sentry** practice from scratch, shipping **reusable dashboards**, response-time monitors, and docs to enhance a previously inefficient maintainer email alert system

Software Developer Intern | Python, FastAPI, Pandas

Jan. 2026 – Apr. 2026

PivoHub

(Remote) Montréal, QC

- Cut customer onboarding from **multiple days to about 1 hour** by shipping an **LLM powered** catalog import pipeline as a production **FastAPI** service, processing supplier catalogs of **1,000+ products**
- Achieved **90%+** mapping accuracy on messy supplier spreadsheets by designing an **LLM driven** engine that auto-detects headers and generates transformation plans, eliminating manual data entry
- Cheapened per-row LLM inference costs by handling **20+** extraction fields in a deterministic regex layer at **95%+** accuracy, reserving embedding similarity for bilingual **29 category** classification

Software Engineer (Incoming co-lead) | TypeScript, Python, Pytest, Pydantic

Sep. 2025 – Present

Waterloo Rocketry

Waterloo, ON

- **Cut Electron app size 50%+** by porting custom Python library (Parsley) from **Python to TypeScript**, eliminating the bundled **runtime** and **dev tool overhead** and unifying debugging into a single language
- Enforced **Pydantic** runtime validation across **60+** unique sensor fields and replaced architecture with an object oriented design, replacing untyped dictionary outputs and **preventing downstream application** crashes
- Implemented high-frequency web telemetry with a bidirectional **WebSocket-ZeroMQ bridge** in **Flask SocketIO**, sustaining **100+ msg/s** to expose low-level rocket bus data to modern web clients
- Mentored new members and helped to **create term goals** and architecture plans for the entire software subsystem

PROJECTS

Pokémon Card Scanner | Python, FastAPI, Next.js, YOLOv8, Docker

[GitHub](#)

- Attained **95%+** identification accuracy across **20,000+** Pokémon cards by building a vision pipeline combining **YOLOv8** detection, **CLIP+FAISS** embedding search, and **OCR** based validation
- Cut multi-card scan latency **50%** with a parallel processing pipeline, deployed and operated in production with a **dockerized FastAPI** backend, **Next.js** frontend, Google OAuth, and **PostgreSQL**

Big Brother | Python, FastAPI, MongoDB, Neo4j

[GitHub](#)

- Won **runner up** at UofT Hacks 2026 by building an autonomous browser agent that **orchestrates 5 LLMs** to execute complex web tasks from natural language, grounded in a **Neo4j knowledge graph** with RAG retrieval
- Reduced inference cost **40%** and average latency from **3.2s to 1.8s (44%)** with **intelligent model routing** and embedding-based semantic caching across the agent fleet

Le Replacer Chrome Extension | JavaScript

[GitHub](#)

- Won **Best Pitch** at Go On Hacks 2025 by building a high-frequency face replacement Chrome extension that modifies dynamically loaded web content using **Shape Detection API**
- **Eliminated severe browser hangs** on media-heavy sites (Instagram/Twitter) by optimizing DOM manipulation algorithms to process high-frequency updates at **60fps** using the **MutationObserver API**